MATH5845

TIME SERIES

GROUP PROJECT REPORT

US Births Dateset Analysis

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Abstract

This study focuses on the US births, aiming at building a time series model for future birth forecasting. The dataset for modelling is US births from 1969 to 1988, translated from daily to monthly. According to the data, two seasonal arima models were built. This study will discuss these models in many fields, especially the fitting situation, residuals of the models and autocorrelation. Furthermore, model selection is also crucial in this study. Particularly, this study will explain how to do the model selection, whether the dataset requires a seasonal model, and how to improve the model performance. Finally, this study will show the forecasting of each model and discuss the results.

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1. Introduction

In this project, US birth dataset was collected from Project MOSAIC, which is funded by the National Science Foundation (NSF), a US government agency that supports scientific research and education. Project MOSAIC is a community composed of educators aiming to integrate the all aspects of quantitative work, including data analysis, mathematical modelling, and computational methods, that students in science, technology, engineering and mathematics (STEM) need for their professional and academic careers [1] .

Although Godahewa et al. [2] do not explicitly justify their choice of the 1969–1988 period in the US births dataset from Zenodo (Monash Time Series Forecasting Repository) [3], the selection may have been influenced by data stability considerations. During the U.S. baby boom era (1946–1964) after World War II, birth rates exhibited irregular and rapidly changing patterns, making long-term trend time-series prediction challenging [4]. By starting in 1969, the dataset likely captures the post–baby boom period, which is characterized by more stable birth rates, less affected by demographic shocks, and more reflective of underlying long-term trend patterns. As a result, this dataset presents lower variability and a clear seasonal pattern, making it suitable for assessing the performance of daily data forecasts [5].

The objectives of this project are:

To analyse long-term trends, seasonal fluctuations (e.g., weekly or yearly cycles), and potential structural changes in U.S. birth rates between 1969 and 1988.

To build statistical time-series models (ARIMA) to capture the underlying temporal dynamics of birth rates by applying appropriate preprocessing techniques such as differencing and seasonal differencing to achieve stationarity.

To determine the best-fitting model for forecasting birth rates by evaluating candidate models using selection criteria (Akaike's Information Criterion (AIC) and residual diagnostics (autocorrelation checks, Ljung–Box test).

To interpret the observed patterns and forecast outcomes in relation to demographic and societal factors during the post-baby boom stabilization period.

This project report contains data description, data preprocessing, results of three SARIMA models and discussion including challenges solved in this study and model evaluation.

2. Data Description

Chosen data is from Monash Time Series Forecasting Repository, describing the number of births in the United State, recording the daily birth in one time series with 7305 lengths, from 01/01/1969 to 31/12/1988. During the dataset, each date index only points to a positive integer value, which is the number of births on the date. And the dataset has no missing value or wrong recording or repeating data.

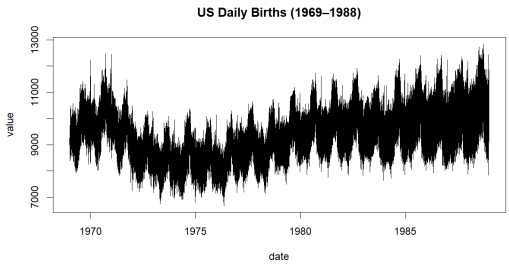

Figure 2.1: Number of US Births from 1969 to 1988

Figure 2.1 visualizes the time series line plot of the US birth dataset. During the plot, the x-lab records the dates, from the first day to the end, and the y-lab records the integer number of birth. According to the Figure 2.1, the number of births was influenced by the time obviously. In 1988, there are 3905292 new births, but in 1973, it is only 3145281. The number of births from the most born year is 24% bigger than the least one.

Moreover, the Figure 2.1 also reveals that the data might be non-stationary. The data has time-varying mean, and the variance at the end part is apparently bigger than the beginning, which means the mean and variance might not be constant. It might need some transformation to make the process stationary.

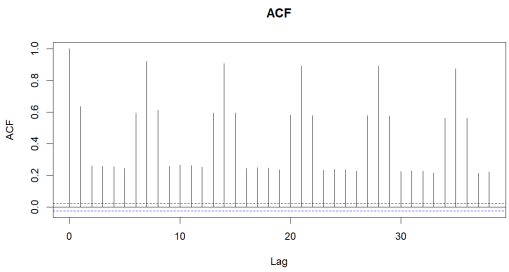

Figure 2.2: ACF of US Births (1969-1988)

Meanwhile, the data might have potential seasonal trends. According to the Autocorrelation Function (ACF) Figure 2.2, the ACF values of first two and last one lags are always higher than the middle four lags during seven lags. Every seven lags repeats this kind of pattern, which means the data has weekly seasonal trends. Additionally, all ACF values are significant, and the figure has a tail off phenomenon.

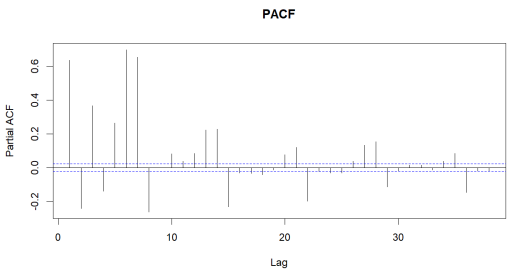

Figure 2.3: PACF of US Births (1969-1988)

The Partial Autocorrelation Function (PACF) Figure 2.3 shows that the PACF values sharply decrease after lag 10 but is still significant, which means the figure still tails off. Combined with the previous result of ACF, there exists a high percentage of probability that the data is not stationary. Besides, it also shows the data might be suitable for an Autoregressive Moving Average (ARMA) model.

3. Methodology

3.1. Tools for Coding

This study relies on the following package from R.

"xts": Used for time series index and resampling.

"tseries": Used for ACF, PACF, and plot of data.

"forecast": Used for Cox-Box transformation, arima and auto.arima modeling with their diagnostic.

3.2. Data Preprocessing

3.2.1. Data Transformation

According to the data description, the original data has apparent seasonal trends weekly and yearly. Preprocessing aims at weaking the seasonal trends, meanwhile controlling the model complexness. Therefore, transforming the data from daily to monthly was used in preprocessing.

The original data contains only a 7304-length series with a time index from 1969-01-01 to 1988-12-30. The monthly transformation is to combine the datapoint having the same year and month index as a new datapoint. After the transformation, the series length is scaled to 240, and the time index becomes 1969-01 to 1988-12.

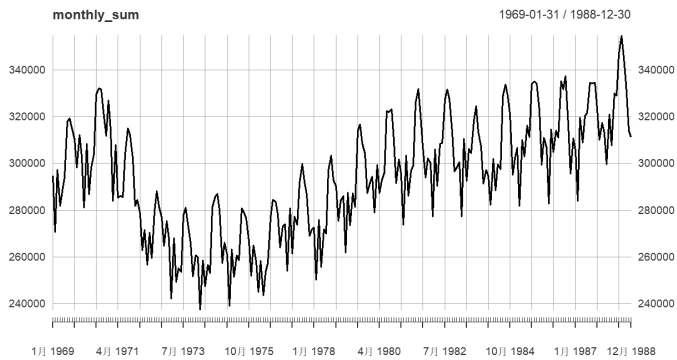

Figure 3.1: Monthly Number of US Births

By the transformation, the weekly effect was removed. The seasonal part alters to contain only yearly trends. And, the scaled data earns better visualizing Figure 3.1, which can expose the trends and fluctuation more obviously. Meanwhile, even though this transformation makes the length of series lose a lot, the remaining length is still acceptable in this study.

3.2.2. Box-Cox Transformation

By analyzing the original data, the following problems were identified from Figure 3.2: (1) Heteroskedasticity: This is mainly manifested in the significant increase in the variance of the birth population over time. (2) Skewed distribution: After aggregating the original data by month, the skewness was found to be negative every month, and the average skewness was -0.777, indicating that the right tail of the data was long and skewed.

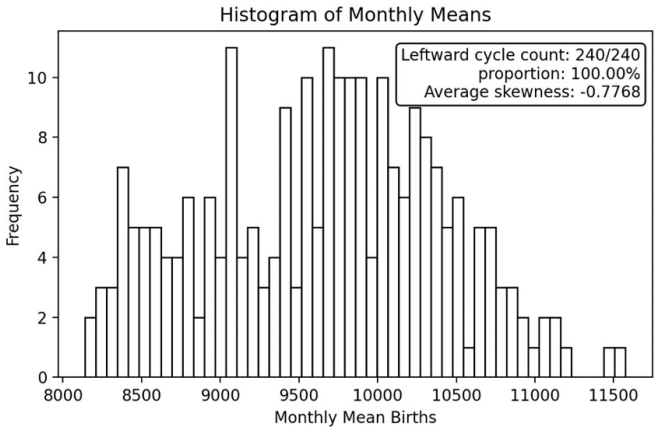

Figure 3.2: Histogram of Monthly Means

These problems violate the basic assumptions of the ARIMA model for normality and homogeneity of variance, and will seriously affect the model fitting effect based on the linear assumption in subsequent modeling. To this end, we introduced the Box-Cox transformation, whose core mechanism is to apply a power function transformation to the original data and adaptively find the optimal parameter λ through the maximum likelihood method to optimize the data distribution characteristics.

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \lambda \neq 0, \\ \ln(y), & \lambda = 0 \end{cases} \tag{1}$$

It nonlinearly maps the original data to a new numerical space and uses a parameter to control the intensity of the data compression or stretching. The value of λ controls the degree and direction of the transformation: when $\lambda > 1$, it enhances the differences in high-value areas; when $0 < \lambda < 1$, it compresses high values and stretches low values to stabilize variance; when $\lambda \to 0$, it performs a near logarithmic transformation to correct skewness; and when $\lambda = 1$, the original data is unchanged.

For our original data, we primarily compress high-value regions and relatively expand low-value regions, thereby suppressing the phenomenon of increasing fluctuations over time. Furthermore, for data with negative skewness, we can stretch the dense left tail and compress the sparse right tail, making the distribution more symmetrical and significantly reducing skewness, bringing it closer to a normal distribution. After completing this transformation, the resulting series exhibits significant improvements in both homogeneity of variance and distribution symmetry, thus better conforming to the basic assumptions of stationarity and normality of models like ARIMA, enhancing the model's fit quality and predictive power.

3.2.3. Differencing

Through analysis, we found that the original data still has the following structural problems: (1) Trend non-stationarity: the data shows a slow upward trend over time, causing the mean to change over time. (2) Seasonal non-stationarity: regular peaks and valleys appear every year, reflecting obvious seasonality. To this end, we used a two-part difference processing method to improve the stationarity of the series:

First, we used the first-order difference, which is specifically to remove the first-order trend term from the original series, so that its mean no longer changes systematically over time, and then transform it into a weak stationary series with a stable mean, limited variance, and autocovariance only related to the lag, which is conducive to improving the fitting effect of the linear model.

To further enhance the stationarity of the series, we also performed seasonal difference processing on the series. Specifically, the first-order difference series is subjected to a delayed difference processing according to the period, and the value of the current period is subtracted from the value of the previous period point, thereby eliminating the structural repetition caused by the periodicity. If there is a stable and repetitive seasonal structure in the series, then the two should be highly similar, and the difference between the two mainly reflects non-seasonal changes. By introducing the joint processing of ordinary and seasonal differences, we effectively converted the original non-stationary series into an approximately stationary series that is more suitable for linear modeling, significantly improving the applicability, explanatory power and predictive accuracy of the model. This ensures that in the subsequent modeling process, we can pay more attention to the random changes in the data rather than systematic seasonal interference, providing a solid foundation for subsequent modelling.

3.3. Models Applied

Based on the stationarity of the preprocessed data and potential seasonal patterns, here are two models would be applied in this study, which are Autoregressive Integrated Moving Average (ARIMA) model and Seasonal ARIMA (SARIMA) model.

3.3.1. ARIMA Model

ARIMA is a statistical modelling approach commonly used for analysing and forecasting time-series data. It integrates three key components: Autoregressive (AR), Integrated (I), and Moving Average (MA). These components collectively capture various characteristics of the time series, such as trend, autocorrelation, and potential seasonal behaviour.

The AR (Autoregressive) component models the current value of the series as a linear combination of its previous values. In essence, it is a regression of the variable on its own lagged (past) terms.

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + Z_t \tag{2}$$

where X_t is stationary, $\phi_1, \phi_2, \ldots, \phi_p$ are constants (with $\phi_p \neq 0$), and $Z_t \sim \mathrm{WN}(0, \sigma^2)$.

Secondly, the Moving Average (MA) component is a time series model that expresses the current value of a variable as a linear combination of past random shocks or error terms. It's the way to capture how past forecast errors influence future values.

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q} \tag{3}$$

where $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$, $\theta_1, \ldots, \theta_q$ are constants ($\theta_q \neq 0$).

The "Integrated (I)" component here refers to the differencing of the time series data. As discussed before, differencing involves subtracting consecutive data points to make the data series stationary, which means that its statistical properties (mean and variance) remain constant over time.

$$Y_t = X_t - X_{t-1} \tag{4}$$

This is referred to as first-order differencing. Higher-order differencing can be applied iteratively if the series remains non-stationary after the first difference. This can be shown as the following format:

$$ARIMA(p, d, q) \tag{5}$$

where parameters p, d, and q represent the order of the AR component, the degree of differencing, and the order of the MA component respectively.

3.3.2. SARIMA Model

However, in some situations, ARIMA may not adequately capture the characteristics of the data. When data presents seasonal patterns recurring at fixed intervals, like monthly sales peaks due to holidays. In such scenarios, SARIMA is particularly designed to account for seasonality as an extension of ARIMA. SARIMA introduces additional seasonal terms to model repeating patterns and cycles, enhancing the model's ability to fit and forecast time series with strong periodic structure.

$$ARIMA(p, d, q) \times (P, D, Q)_s \tag{6}$$

where non-seasonal parameters p,d,q have been explained before, whereas additional seasonal parameters P, D, Q, s represent the order of seasonal AR component, the order of seasonal differencing, the order of seasonal MA component and seasonal period

In the SARIMA model, seasonal AR and MA terms can forecast future values by using past data values at lags that are multiples of the seasonal period s (the span of the seasonality). For example, if the model is used to capture patterns that repeat annually in monthly data, the value of s would be set to 12.

3.4. Models Evaluation

Model evaluation in this study primarily uses the Akaike's Information Criterion (AIC) as metric, constructed with the following equation:

$$AIC = \log(\hat{\sigma}^2) + \frac{n + 2k}{n} \tag{7}$$

where the $\hat{\sigma}^2$ is the maximum likelihood estimator of the residual variance, n is the number of observations and k is the number of parameters.

The best model chosen requires the lower AIC value than all other models. The main reason is that the residual variance $\log(\hat{\sigma}^2)$ means the variance which could not be explained by the model. Additionally, the penalize term $\frac{n+2k}{n}$ restricts the complexity of the model, which prevents overfitting.

3.4.1. Diagnostic Tests

The diagnosis of this study would rely on the diagnostic plot, including four important parts.

Standardized Residual: The plot shows the standardized residual change with the time change. The mean and variance can be observed from this plot, which can help to recognize whether the residual $\hat{e} \sim N(0, \sigma^2)$.

ACF of Residual: The plot shows whether each residual is correlated with the residuals after several lags. It can be observed whether the residuals are independent and identically distributed.

Q-Q Plot: The plot shows whether residual quantiles follow the normal quantiles. Normality can be observed from this plot. P-value for Ljung-Box: This plot reflects the significance of the hypothesis test, where the null hypothesis is the ACF $\rho_1 = \rho_2 = ... = \rho_m = 0$.

4. Results

4.1. ARIMA$(0,1,1)(0,1,1)_{12}$

After applying auto.arima function with default settings on the monthly data, ARIMA$(0,1,1)(0,1,1)_{12}$ model can be obtained. The following summary generated from this model is listed below:

```
Series: seasonal_monthly_sum_transformed
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
          ma1     sma1
      -0.3157  -0.7659
s.e.   0.0635   0.0486

sigma^2 = 0.0002653:  log likelihood = 612.16
AIC=-1218.33   AICc=-1218.22   BIC=-1208.06
```

Figure 4.1: Summary of ARIMA$(0,1,1)(0,1,1)_{12}$

$$X_t = (1 - 0.3157B)(1 - 0.7659B^{12})Z_t \tag{8}$$

And the diagnostic tests are shown below:

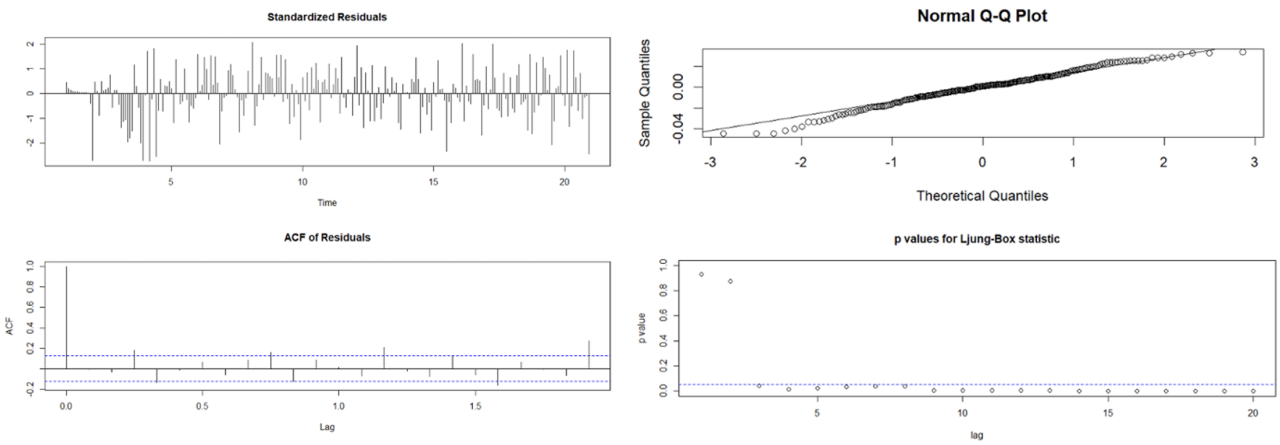

Figure 4.2: Diagnostic Tests of ARIMA$(0,0,1)(0,1,1)_{12}$

From the diagnostic tests, most of the residuals stay within -2 to 2, and there is no clear pattern observed in the standardized residuals. This behaves like white noise, indicating that the residuals are approximately uncorrelated in the time domain with relatively stable variance.

The residuals on the left tail deviate from the 45-degree line in the Normal Q–Q plot, indicating slight deviations from normality. However, the overall distribution of residuals remains reasonably close to normal, indicating that the model is acceptable in residual normality.

However, the ACF plot of residuals shows about 5 significant values excess the blue lines and others are very small, indicating that the residual autocorrelation exists at few specific lags. Thus, the model requires further improvement to reduce the amount of these significant values.

Moreover, the p values for Ljung-Box statistic are almost all lower than 0.05 after lag 2. This rejects the null hypothesis of no autocorrelation, which indicates the existence of autocorrelation in the residuals at higher lags. Therefore, the model fails to fully capture all time-dependent structure in the data and cannot fit the data well.

4.2. ARIMA$(2,1,5)(1,1,1)_{12}$

Therefore, the model with manual set parameters was used to capture delayed dependencies in the time series and reduce residual autocorrelation. The model was set to be ARIMA$(2,1,5)(1,1,1)_{12}$, the selection of parameters including p, q, P, Q values here will be further explained in Section **??**.

```
ARIMA(2,1,5)(1,1,1)[12]

Coefficients:
          ar1      ar2     ma1     ma2      ma3      ma4     ma5     sar1     sma1
      -1.1615  -0.9979  0.9572  0.7686  -0.1959  -0.0137  0.0166  -0.0948  -0.7209
s.e.   0.0047   0.0032  0.0700  0.0967   0.1104   0.1001  0.0716   0.0877   0.0646

sigma^2 = 0.0002218:  log likelihood = 633.14
AIC=-1246.28   AICc=-1245.26   BIC=-1212.03
```

Figure 4.3: Summary of ARIMA$(2,1,5)(1,1,1)_{12}$

$$(1+1.1615B+0.9979B^2)(1+0.0948B^{12})X_t = (1+0.9572B+0.7686B^2-0.1959B^3-0.0137B^4+0.0166B^5) \times (1-0.7209B^{12})Z_t \tag{9}$$

From the summary of the $ARIMA(2,1,5)(1,1,1)_{12}$ model, it has clearly lower AIC/ AICc values compared to the $ARIMA(0,1,1)(0,1,1)_{12}$ model, indicating a better balance between goodness-of-fit and model complexity.

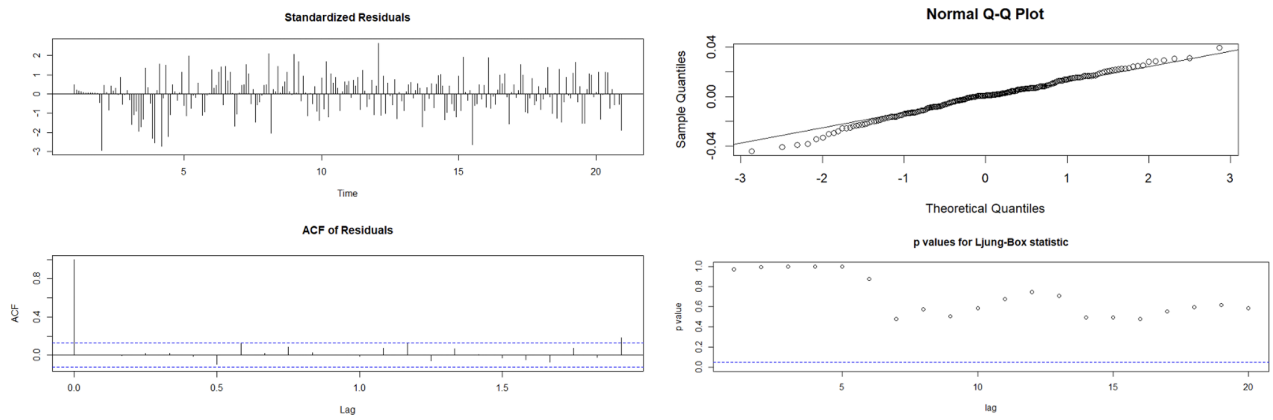

Figure 4.4: Diagnostic Tests of ARIMA$(2,1,5)(1,1,1)_{12}$

Compared this diagonostic tests with $ARIMA(0,1,1)(0,1,1)_{12}$ model above, the standardized residuals and Normal Q–Q plot show no obvious difference, indicating that this model maintains stable variance and is acceptable in residual normality. However, the number of significant spikes in the ACF of residuals is reduced from 5 to only 1, indicating that residual autocorrelation is almost entirely eliminated. Moreover, all p-values from the Ljung–Box test are greater than 0.05, which is an important improvement, as it means this model cannot reject the null hypothesis of no autocorrelation. Therefore, these results indicate that the model effectively captures all time-dependent structure in the data and provides a good fit with data.

4.2.1. Simplification of ARIMA$(2,1,5)(1,1,1)_{12}$

Moreover, by considering the s.e. and coefficients, the model could be further simplified because they are unsignificant, which are $\theta_3, \theta_4, \theta_5$ and Φ_1, the simplified model $ARIMA(2,1,2)(0,1,1)_{12}$ is shown below:

```
ARIMA(2,1,2)(0,1,1)[12]

Coefficients:
          ar1      ar2     ma1     ma2     sma1
      -1.1626  -0.9973  1.1262  1.0000  -0.7892
s.e.   0.0050   0.0033  0.0209  0.0206   0.0475

sigma^2 = 0.0002228:  log likelihood = 629.53
AIC=-1247.06   AICc=-1246.68   BIC=-1226.51
```

Figure 4.5: Summary of ARIMA$(2,1,2)(0,1,1)_{12}$

$$(1+1.1626B+0.9973B^2)X_t = (1+1.1262B+B^2)(1-0.7892B^{12})Z_t \tag{10}$$

4.3. ARIMA$(4,1,0)(0,1,1)_{12}$

After the previous two models' results, a key adjustment on the auto.arima function has been identified, which is changing the default settings of this function. Both stepwise and approximation are set to FALSE to generate the model, which gives the model better accuracy. Because stepwise = FALSE means the model will use an exhaustive search, and approximation = FALSE means the exact model fitting is used, it can avoid local optimal solutions and give a better fit model.

```
ARIMA(4,1,0)(0,1,1)[12]

Coefficients:
          ar1      ar2     ar3      ar4     sma1
      -0.2857  -0.1270  0.1407  -0.0977  -0.7633
s.e.   0.0684   0.0699  0.0690   0.0664   0.0485

sigma^2 = 0.0002552:  log likelihood = 618.29
AIC=-1224.58   AICc=-1224.2   BIC=-1204.03
```

Figure 4.6: Summary of ARIMA$(4,1,0)(0,1,1)_{12}$

$$(1+0.2857B+0.1270B^2-0.1407B^3+0.0977B^4)X_t = (1-0.7633B^{12})Z_t \tag{11}$$

In this model summary, ARIMA$(4,1,0)(0,1,1)_{12}$ shows similar but slightly higher AIC/AICc than ARIMA$(2,1,5)(1,1,1)_{12}$, but still much lower than the first model ARIMA$(0,1,1)(0,1,1)_{12}$, also generated from the auto.arima function, which proves that these adjustments to the function are valid and effective in improving model performance.

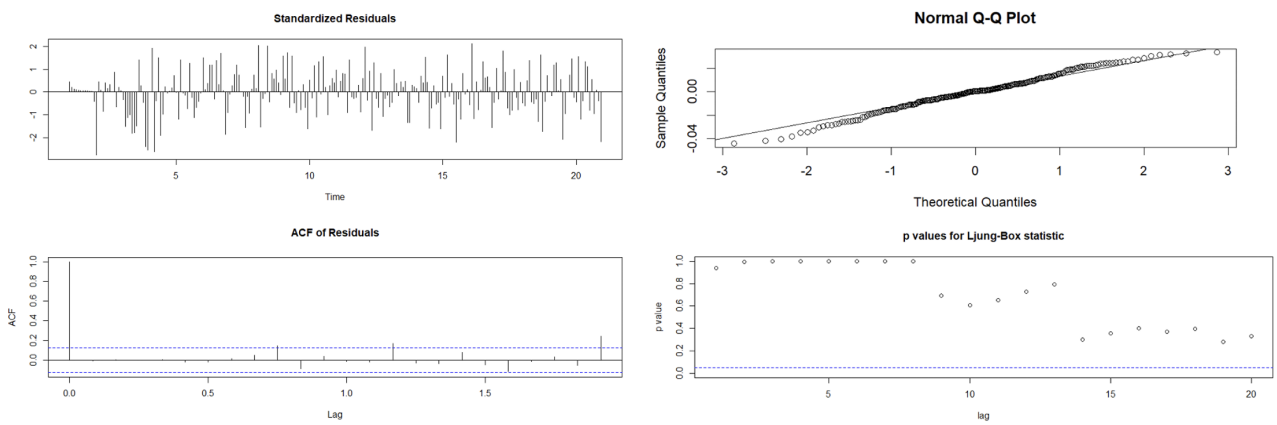

Figure 4.7: Diagnostic Tests of ARIMA$(4,1,0)(0,1,1)_{12}$

Compared with the manually set model $ARIMA(2,1,5)(1,1,1)_{12}$, there is no obvious difference in the standardized residuals, Normal Q–Q plot, or p-values from the Ljung–Box statistic. However, the number of significant values in the ACF increases from 2 to 3, indicating that this model has slightly more residual autocorrelation at higher lags. Overall, its performance is better than the earlier $ARIMA(0,1,1)(0,1,1)_{12}$ model but slightly worse than the $ARIMA(2,1,5)(1,1,1)_{12}$ model. This is due to the limitations of the auto.arima function settings, as the max.p and max.q values are still at their default of 5, which constrains the orders of the parameters and reduces the search space

for candidate models. In contrast, the $ARIMA(2,1,5)(1,1,1)_{12}$ model has seven AR/MA parameters in total, enabling a better fit.

5. Discussion

5.1. Lambda Value in Box-Cox Transformation

In the Box–Cox transformation applied during the modelling process, the parameter λ was set to 0. As defined in Section 3.2.2, different values of λ correspond to specific transformations: when $\lambda = 1$, the transformation has little effect beyond a constant shift; when $\lambda = 0.5$, it reduces to a square root transformation; and when $\lambda = -1$, it becomes a reciprocal transformation, which may excessively distort relationships. Setting $\lambda = 0$ applies a natural logarithmic transformation to the dataset. Given that the dataset contains only positive values and the variance increasing alongside the mean, therefore the log transformation is appropriate here to stabilize the variance.

5.2. Manual Model Selection

The needed manual model requires more parameters than the default auto.arima model. Furthermore, the manual model had better be able to get a lower AIC as possible.

To achieve this, suppose the number of parameters is k, the number of p in the manual ARIMA model was a sequence from 0 to k, and q was k-p. When the number of k=7, most of the p-values for Ljung-Box statistic were significant in observation. Therefore, the p+q was set to be 7. Meanwhile, considering the seasonal trend of the data, a basic seasonal component $(p,d,q)(1,1,1)_{12}$ is used to eliminate that.

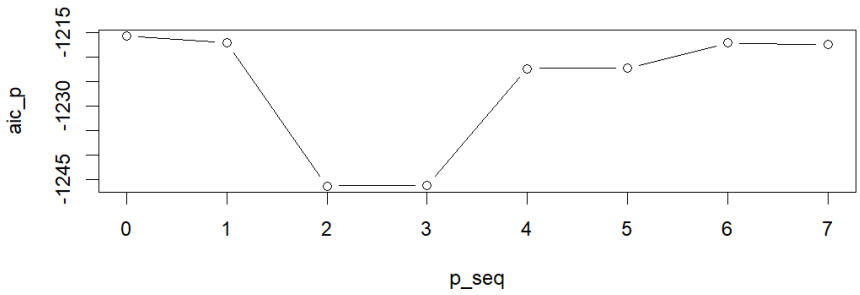

Figure 5.1: AIC Trend

The visualization of the change in AIC with p and q is shown in Figure (). When p=2 or p=3, the AIC is visibly lower than other models. That is why the manual model was set to be ARIMA$(2,1,5)(1,1,1)_{12}$.

5.3. Local Optimal Solution of auto.arima

The R function "auto.arima" in the "forecast" package uses AICc as the standard to compare each model. As $AICc = AIC + \frac{2k(k+1)}{n-k-1}$, once the number n was not large, the penalty to the new parameters added comparatively would be larger, so the auto.arima would tend to choose small models. Furthermore, AICc gives more penalty to the newly added parameters, which leads to the potential risk of underfitting.

Except that, auto.arima uses approximation and stepwise by default. Since there might exist a possible overlarge penalty to parameters, adding a new parameter might be hard to reduce AICc. Then, limited by the stepwise algorithm, once a new parameter added cannot reduce AICc, the auto.arima would stop checking for more models, which might lead to falling into a local optimal solution.

In this dataset, because the data was transformed from daily to monthly, the data length decreased a lot. The remaining length might cause restrictions on the selection of large models. That could be the reason why the default auto.arima chose an underfitting model.

5.4. Effect of Seasonal

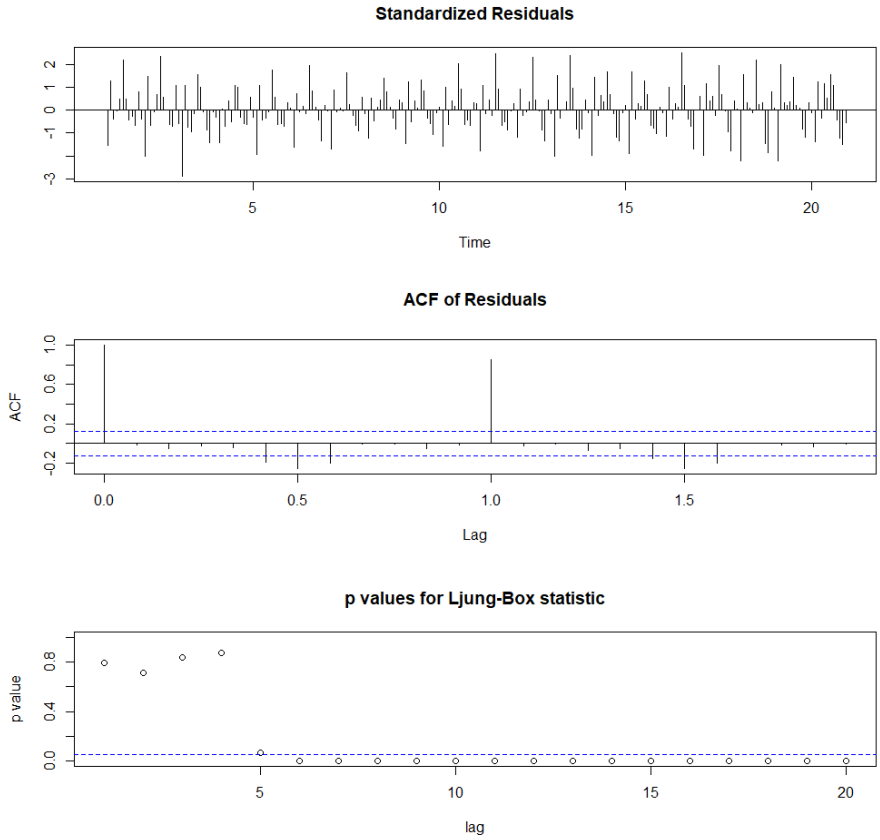

Figure 5.2: Diagnostic Tests of ARIMA(4,1,0)

To test whether the seasonal method truly works, a non-seasonal arima model with same (p,d,q) was established. Figure 5.2 shows the diagnostic plot of ARIMA(4,1,0) model, used as a comparison of the model ARIMA$(4,1,0)(0,1,1)_{12}$.

According to the ACF residual plot, most of the autocorrelations are under the p-value line. However, the ρ_{12} in ACF plot is very high, which means each residual is highly correlated with its lag 12. This also reflects in the Ljung-Box plot. Once the lag includes ρ_{12}, there is strong evidence to reject the null hypothesis. These all indicate that the model needs a seasonal difference on s=12.

When giving a seasonal component to the ARIMA(4,1,0), the ρ_{12} in ACF plot are below the blue line, and the Ljung-Box plot shows the p value starts to be all significant over 20 lags. That means the seasonal model truly works.

5.5. Model Performance

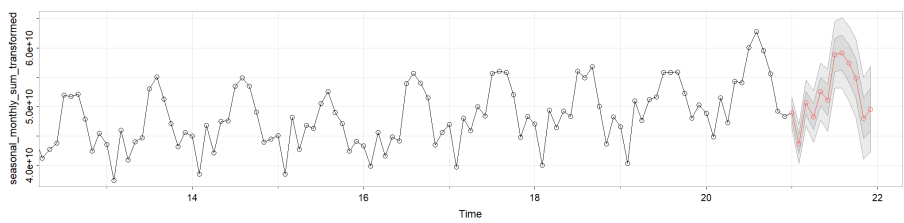

Figure 5.3: Prediction of ARIMA$(2,1,5)(1,1,1)_{12}$ in the future 1 year

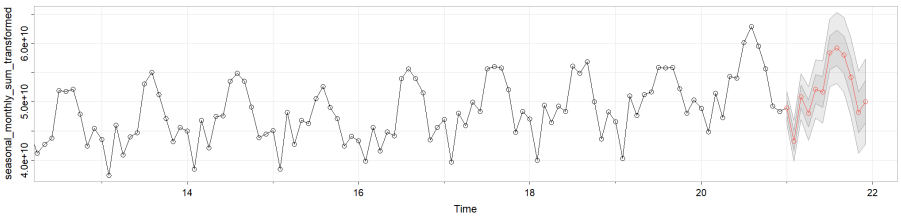

Figure 5.4: Prediction of ARIMA$(4,1,0)(0,1,1)_{12}$ in the future 1 year

Figure 5.3 and 5.4 show 12 forecasts of the manual model ARIMA$(2,1,5)(1,1,1)_{12}$ and the auto model ARIMA$(4,1,0)(0,1,1)_{12}$. These two models have similar forecasts. These forecasts catch the trends of the previous births. And the change of the prediction interval matches the change of points. The model does not forecast by only on linearity. Therefore, the performance of these models is not bad.

6. Conclusion

This report analysed U.S. monthly birth data from 1969 to 1988 using SARIMA modelling to capture long-term trends, seasonal effects, and short-term dependencies. After comprehensive preprocessing, the original daily birth series was aggregated to a monthly frequency, effectively removing the weekly seasonal component. A Box–Cox transformation was then applied to stabilize variance, followed by a combination of regular and seasonal differencing to eliminate long-term trends and annual seasonality, thereby achieving stationarity in both mean and variance. These steps ensured that the transformed monthly series satisfied the assumptions required for fitting SARIMA models.

From the comparison of these models, the manually set model ARIMA$(2,1,5)(1,1,1)_{12}$ achieved the best overall performance, with lower AIC/AICc values, minimal residual autocorrelation, which is better than both ARIMA$(0,1,1)(0,1,1)_{12}$ auto model with default settings and the adjusted ARIMA$(4,1,0)$ $(0,1,1)_{12}$ model. The analysis also demonstrated that seasonal differencing at s=12 was essential for eliminating significant lag-12 autocorrelations.

Finally, the forecasting of the manual model has similar mean and variance to the auto model. However, the forecasting of the auto model has a softer shape. This might mean the model ARIMA$(4,1,0)$ $(0,1,1)_{12}$ is more suitable for generalization. If there is a test set about the number of US births over 1988, the auto model should work better on the test set.

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